a **Study design: four parallel tracks converge**

Bulk RNA
TARGET-OS
n = 88
+ GEO cohorts (planned)

Single-cell
GSE152048
GSE162454

ICB cohorts
4 pan-cancer
transfer only

Cell lines/drugs
DepMap
GDSC2

public data only

A **Clinical**
OS/EFS, metastasis
Chemo response
Nomogram

B **Molecular**
IFN-γ → STAT1
CXCR3 vs ACKR3
Pathways (GSEA)

C **TME**
Deconvolution
Single-cell map
Cell–cell signalling

D **Immunotherapy**
TIDE / IPS
ICB transfer
Drug sensitivity

Convergence: CXCL11 compartment balance
$ABS = R_{imm} - (R_{tum} + S_{endo}) / 2$
modules defined in single-cell data · interpretable ML (nested CV, SHAP)

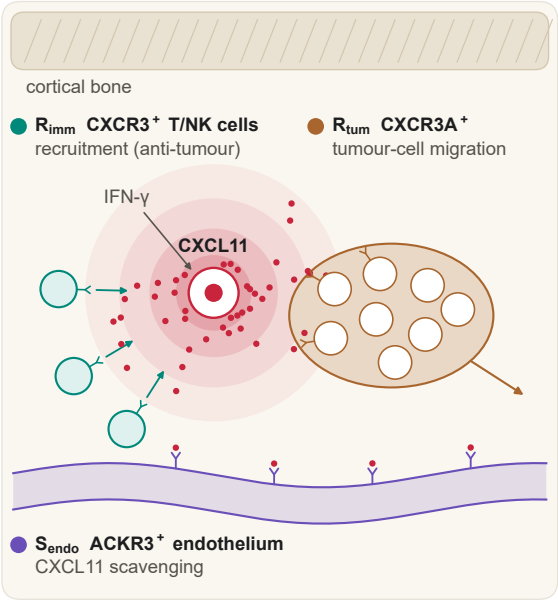
Prognostic stratification

Immunotherapy relevance

Open atlas and pipeline

Open and reproducible by design
one-command Snakemake pipeline · fixed seeds · locked environments
every reported number traced to a pipeline output
code and atlas to be archived with a DOI on release

b **Three-compartment hypothesis**



Hypothesis: a higher balance score (ABS)
marks an immune-favourable niche

